

1. Questions

Two series I and II are given below:

I: a, b, c, 60, 85, 115, 150

II: (a + 5), (b + 25), (c + 60), d, e, f, 0

Both the series follow different patterns, and $abc = 2n(d + e + f)$, then find the value of 'n'.

- a. 25
- b. 40
- c. 20
- d. 15
- e. 10

2. Questions

The series given below is Series I, which follows a certain pattern. Find the average of the missing terms in Series II, which follows the same logic as Series I.

I: (a - b), (c - d), (c + d), (a+b), 132, 232

II: (a + b - c + d), ?, ?, ?, ?

Notes:

i). The difference between 1st and 4th terms of series I is 56.

ii). $7a = 10b$ and $c = 3d$

- a. 102
- b. 112
- c. 98
- d. 108
- e. 105

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3. Questions

The given series in which all the terms are unknown and can be found from the given information.

A, B, C, D, E, F, G, H

A = Sum of the smallest single-digit prime number and the largest single-digit prime number.

B = Sum of roots of the equation $(x - 10)^2 - 1 = 0$

C = Second smallest prime number * 11.

D = '1' more than a perfect square of an integer and also a multiple of 10.

E = The sum of the digits of the number is 10, while the 1st digit is 4 more than the 2nd digit.

F = $13n$ ('n' is the largest single-digit perfect cube of an integer.).

E > D > C

Find the sum of the values of G and H.

- a. 343
- b. 345
- c. 291
- d. 307
- e. 311

4. Questions

Find the HCF of $(G + 5)$, $(H + 2)$, and $(C + D - 8)$.

- a. 10
- b. 20
- c. 15
- d. 25
- e. 30

5. Questions

The two series, I and II, are given below. Series I has one missing term 'P' while series II has two wrong terms which are to be replaced by correct terms 'Q and R' ($Q < R$).

Quantity I: The value of 'P'.

Quantity II: The value of 'Q' and 'R'.

Quantity I __ **Quantity II**.

Which of the following symbol can fill the above blank?

Series I: 11, 24, P, 304, 1525, 9156

Series II: 3, 8, 15, 26, 39, 55, 70, 98

- a. $\leq$
- b. $\geq$
- c. $<$
- d. $>$
- e. $=$

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6. Questions

Given below are two equations, I and II. Solve the equations and answer the questions.

I: $(a + b)x^2 + ax + (b + 19) = 0$

II: $2y^2 - 19y - (5P - 2) = 0$

Consider the statements given below:

- a). The coefficient of 'x' and the constant term in equation I are equal.
- b). One root of equation II is 11, while the coefficient of x^2 in equation I is 'P'.

Find the correct relationship between 'x' and 'y'.

- a. $x > y$
- b. $x \geq y$
- c. $x \leq y$
- d. $x = y$ or relationship between 'x' and 'y' cannot be determined.
- e. $x < y$

7. Questions

Which of the following number/equation must be added to get the real roots of equation I?

A: $11x - 4$

B: $x^2 - 11x - 16$

C: $x + 3$

- a. Only A and B
- b. Only A
- c. Only B
- d. Only B and C
- e. Only A and C

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8. Questions

Three quadratic equations are given below. Solve them to determine the answer the following questions.

I: $Px^2 + 13x - 7 = 0$

II: $y^2 - Qy - 88 = 0$

III: $Rz^2 + 24z - R = 0$

- a). P, Q, and R are prime integers.
- b). Sum of the values P, Q, and R is 10.
- c). $P < Q < R$.

Find the product of the largest roots of all three equations.

- a. $1(3/10)$
- b. $1(2/15)$
- c. $1/(1/5)$
- d. $1(1/10)$
- e. $9/10$

9. Questions

Find the LCM of ' $P * Q * R$ ', ' $4P + 2Q - 1$ ', and ' $5(Q + R)$ '.

- a. 90
- b. 1560
- c. 80
- d. 150
- e. 60

10. Questions

If $(a - P)(b - Q)(c - R) = 1001$, then find the sum of values of a, b, and c (a, b, and c are more than 3, and $a < b < c$).

- a. Cannot be determined
- b. 37
- c. 43
- d. 38
- e. 41

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11. Questions

The Total number of balls faced by Rohit in a match is 60 less than the total number of balls faced by Rahul in the same match. If together they scored 100 runs in the match and the strike rate of Rohit is $15/4$ times the strike rate of Rahul, then which of the following can be the runs scored by Rohit and Rahul, respectively?

Strike Rate = (Runs scored/balls faced) * 100.

Note: Strike rates of both the batsman are integer values.

I: 30 and 70

II: 60 and 40

III: 50 and 50

- a. Only I and II
- b. Only I
- c. All I, II, and III
- d. None
- e. Only II

12. Questions

A boat can go 50 (P) km upstream in 4 hours while going 40 (Q) km downstream in 2.5 hours. Also, the boat goes 30(R) km downstream and comes back to the same point and takes a total of 4.5 hours. If the ratio of the speed of the boat in still water to the speed of the stream is 3:1, respectively, then find the correct order of the values associated with the variables P, Q, and R in the bracket.

- a. P-R-Q
- b. Q-R-P
- c. Q-P-R
- d. P-Q-R
- e. R-P-Q

13. Questions

Printer A can print 120 pages in 7.5 min, while printer B can print 90 pages in 4.5 min. The efficiency of another printer C, is in between the efficiencies of printers A and B, and printer D is less efficient than printer A. If all the printers print at least one page, then what can be the difference between the total number of minimum and maximum number of pages printed by printers C and D together in 1 min?

- a. 50
- b. 20
- c. 16
- d. Cannot be determined
- e. 60

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14. Questions

The compound interest earned on an amount when invested at an annual compound interest rate for 2 years is ₹400 more than the simple interest earned on the same amount at the same rate for

the same period of time. The compound interest will be ₹5600 more when invested at twice the original annual compound interest rate while keeping the simple interest rate the same. Find the principal amount.

- a. ₹5000
- b. ₹20000
- c. ₹15000
- d. ₹10000
- e. ₹30000

15. Questions

The difference between the speed of a car and a bike is 15 km/h. When both started from the same point in opposite directions, then the distance between them after 1.2 hours becomes 126 km, then which of the following statement(s) given below is/are possibly TRUE?

I: The distance travelled by car in 2 hours is 100 km.

II: When both are 12 km apart initially and the car is chasing the bike in the same direction, then it will catch the bike in 48 min.

III: The distance travelled by bike in 1.5 hours is 75 km.

- a. Only II
- b. Only I and II
- c. Only I and III
- d. None
- e. Only II and III

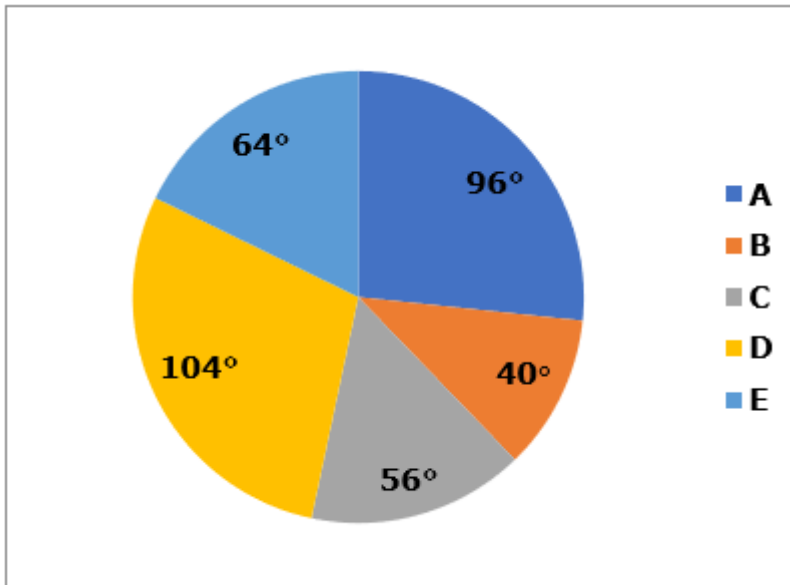
16. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of passengers who travelled by five trains, A, B, C, D, and E, on a particular day.

The pie chart given below shows the degree distribution of the number of passengers who travelled by each of the given trains.

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The table given below shows the ratio of male passengers to female passengers in each of the given trains.

Train	The ratio of male to female passengers
A	7: 5
B	2: 3
C	11: 24
D	9: 4
E	3: 5

Notes:

- 1: The difference between the number of female passengers in trains B and C is 45.
- 2: The total number of passengers = The number of (male + female) passengers
- 3: The number of male passengers = The number of (male adults + male children)
- 4: The number of female passengers = The number of (female adults + female children)

The ratio of the number of male adults to that of male children in train C is 8:3, and the ratio of the number of female adults to that of female children in train C is 13:11. If the ratio of the total number of adults to the total number of children in train C is M:N, then find which of the following relations between M and N is not correct.

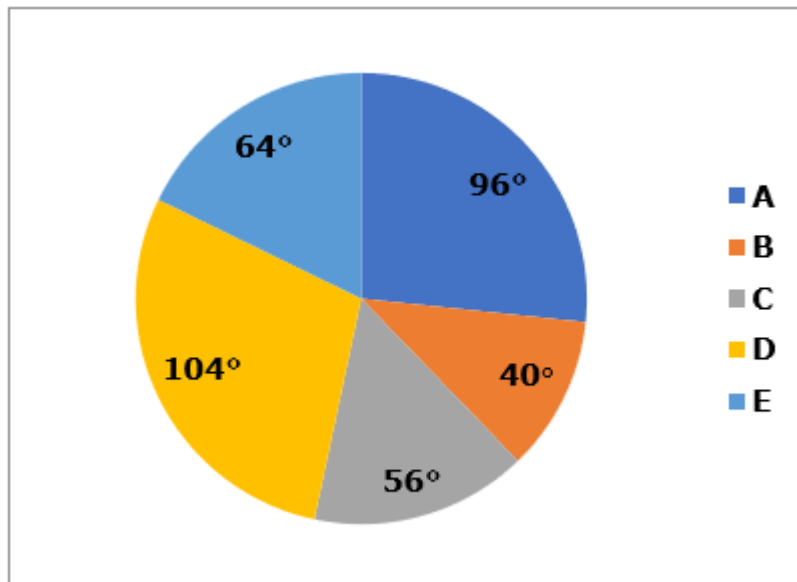
- a. $5M = 11N - 5$
- b. $7M - 4N = 13$
- c. $M^3 - N^3 = 21$
- d. Both (a) and (c)
- e. Both (b) and (c)

17. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of passengers who travelled by five trains, A, B, C, D, and E, on a particular day.

The pie chart given below shows the degree distribution of the number of passengers who travelled by each of the given trains.



The table given below shows the ratio of male passengers to female passengers in each of the given trains.

Train	The ratio of male to female passengers
A	7: 5
B	2: 3
C	11: 24
D	9: 4
E	3: 5

Notes:

- 1: The difference between the number of female passengers in trains B and C is 45.
- 2: The total number of passengers = The number of (male + female) passengers
- 3: The number of male passengers = The number of (male adults + male children)
- 4: The number of female passengers = The number of (female adults + female children)

In train E, the ratio of male adults to male children is 11: 4 and the ratio of female adults to female children is 18: 7. If the ticket for adults is ₹80 per adult and that of children is ₹40 per child, then find that the total amount received by selling tickets for adults in train E is approximately what percent more than that received by selling tickets for children in train E?

- a. 323%
- b. 427%

- c. 475%
- d. 345%
- e. 393%

18. Questions

When the numbers of male passengers in each of the given trains are described in a circle, the degree distribution of the number of male passengers in train B is P° . When the numbers of female passengers in each of the given trains are described in a circle, the degree distribution of the number of female passengers in train D is Q° . Find the ratio of P to Q.

- a. 109: 132
- b. 209: 232
- c. 109: 232
- d. 209: 132
- e. None of these

19. Questions

If $41\frac{2}{3}\%$ of the total number of passengers in train A are children and the ratio of male children to female children in train A is 12: 13, then find the ratio of the number of male adults to the number of female adults in train A.

- a. 5: 3
- b. 23: 12
- c. 23: 15
- d. None of these
- e. 5: 4

20. Questions

A series is formed, whose:

1st term = The number of male passengers in train B.

2nd term = The number of male passengers in train C.

3rd term = The number of male passengers in train E.

4th term = The number of male passengers in train C + The number of female passengers in train D.

5th term = The total number of passengers in train D + 150.

Find the 6th term of the series.

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- a. 1445
- b. 1225
- c. 1895
- d. 1675
- e. 1755

Explanations:

1. Questions

Answer: C

In series I:

Difference between 4th and 5th term = $85 - 60 = 25$

Difference between 5th and 6th term = $115 - 85 = 30$

Difference between 6th and 7th term = $150 - 115 = 35$

So, 3rd term = $60 - 20 = 40 = c$

2nd term = $40 - 15 = 25 = b$

1st term = $25 - 10 = 15 = a$

Now, series II:

=> $(a + 5), (b + 25), (c + 60), d, e, f$

=> 20, 50, 100, d, e, f

The logic is:

$$20 * 2.5 = 50$$

$$50 * 2 = 100$$

$$100 * 1.5 = 150 = d$$

$$150 * 1 = 150 = e$$

$$150 * 0.5 = 75 = f$$

$$75 * 0 = 0$$

Now,

$$abc = 2n(d + e + f)$$

$$15 * 25 * 40 = 2n(150 + 150 + 75)$$

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$$n = 7500/375$$

$$n = 20$$

2. Questions

Answer: A

Series I:

$$(a - b), (c - d), (c + d), (a + b), 132, 232$$

Given: $7a = 10b \Rightarrow b = 0.7a$ and $c = 3d$

$$\Rightarrow (a - 0.7a), (3d - d), (3d + d), (a + 0.7a), 132, 232$$

$$\Rightarrow 0.3a, 2d, 4d, 1.7a, 132, 232$$

The difference between 1st and 4th terms of series I is 56.

$$1.7a - 0.3a = 56$$

$$1.4a = 56$$

$$a = 40$$

$$0.3a = 12$$

$$1.7a = 68$$

$$\Rightarrow 12, 2d, 4d, 68, 132, 232$$

$$\text{Difference between 4th and 5th term} = 132 - 68 = 64 = 8^2$$

$$\text{Difference between 5th and 6th term} = 232 - 132 = 100 = 10^2$$

This means the logic of series I could be the difference of the square of even integers.

So,

$$232 - 10^2 = 132$$

$$132 - 8^2 = 68 = (a + b)$$

$$68 - 6^2 = 32 = (c + d)$$

$$32 - 4^2 = 16 = (c - d)$$

$$16 - 2^2 = 12 = (a - b)$$

$$c = (32 + 16)/2 = 24 \text{ and } d = (32 - 16)/2 = 8$$

Now,

Series II:

$$(a + b - c + d), ?, ?, ?, ?$$

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52, ?, ?, ?, ?

The logic is:

$$52 + 2^2 = 56$$

$$56 + 4^2 = 72$$

$$72 + 6^2 = 108$$

$$108 + 8^2 = 172$$

$$\text{Required average} = (56 + 72 + 108 + 172)/4$$

$$= 408/4$$

$$= 102$$

3. Questions

A, B, C, D, E, G, H

Smallest single-digit prime number = 2

Largest single-digit prime number = 7

$$A = 2 + 7 = 9$$

$$(x - 10)^2 - 1 = 0$$

$$(x - 10)^2 = 1$$

$$(x - 10) = \pm 1$$

$$x = 11 \text{ and } 9$$

$$B = 11 + 9 = 20$$

The second smallest prime multiple = 3

$$C = \text{Second smallest prime number} * 11 = 3 * 11 = 33$$

Perfect square of integers = 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121,

Possible values of 'D' = 2, 5, 10, 17, 26, 37, 50, 65, 82, 101, 122,

Let solve for 'E' first to check the range of values of 'D'.

$$1^{\text{st}} \text{ digit} = (10 + 4)/2 = 7$$

$$2^{\text{nd}} \text{ digit} = (10 - 4)/2 = 3$$

$$E = 73$$

So, $E > D > C$

$$73 > D > 33$$

Since, 'D' is 1 more than the square of integers and also a multiple of 10.

Out of all possible values of 'D', only 50 satisfies the condition.

So, $D = 50$

$n = \text{Largest single digit perfect cube of an integer} = 8$

$F = 13n = 8 * 13 = 104$

So, series: A, B, C, D, E, F

$\Rightarrow 9, 20, 33, 50, 73, 104$

The logic is:

So, $G = 145$ and $H = 198$

Answer: A

Sum of values of G and H = $145 + 198 = 343$

4. Questions

A, B, C, D, E, G, H

Smallest single-digit prime number = 2

Largest single-digit prime number = 7

$A = 2 + 7 = 9$

$(x - 10)^2 - 1 = 0$

$(x - 10)^2 = 1$

$(x - 10) = \pm 1$

$x = 11$ and 9

$B = 11 + 9 = 20$

The second smallest prime multiple = 3

$C = \text{Second smallest prime number} * 11 = 3 * 11 = 33$

Perfect square of integers = 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121,

Possible values of 'D' = 2, 5, 10, 17, 26, 37, 50, 65, 82, 101, 122,

Let solve for 'E' first to check the range of values of 'D'.

$1^{\text{st}} \text{ digit} = (10 + 4)/2 = 7$

$2^{\text{nd}} \text{ digit} = (10 - 4)/2 = 3$

$E = 73$

So, $E > D > C$

$$73 > D > 33$$

Since, 'D' is 1 more than the square of integers and also a multiple of 10.

Out of all possible values of 'D', only 50 satisfies the condition.

$$\text{So, } D = 50$$

$$n = \text{Largest single digit perfect cube of an integer} = 8$$

$$F = 13n = 8 * 13 = 104$$

So, series: A, B, C, D, E, F

$$\Rightarrow 9, 20, 33, 50, 73, 104$$

The logic is:

$$\text{So, } G = 145 \text{ and } H = 198$$

Answer: D

$$(G + 5) = 145 + 5 = 150 = 2 * 3 * 5^2$$

$$(H + 2) = 198 + 2 = 200 = 2^3 * 5^2$$

$$(C + D - 8) = 33 + 50 - 8 = 75 = 3 * 5^2$$

$$\begin{aligned} \text{HCF of } (G + 5), (H + 2), \text{ and } (C + D - 8) &= \text{HCF of } 150, 200, \text{ and } 75 = 5^2 \\ &= 25 \end{aligned}$$

5. Questions

Answer: B

Series I: 11, 24, 75, P, 1525, 9156

The logic is:

$$11 * 2 + 2 = 24$$

$$24 * 3 + 3 = 75 = P$$

$$75 * 4 + 4 = 304$$

$$304 * 5 + 5 = 1525$$

$$1525 * 6 + 6 = 9156$$

$$\text{So, } P = 75$$

Series II: 3, 8, 15, 26, 39, 55, 70, 98

The logic is (Difference of prime numbers):

$$3 + 5 = 8$$

$$8 + 7 = 15$$

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$$15 + 11 = 26$$

$$26 + 13 = 39$$

$$39 + 17 = 56 \text{ (Not 55)} \Rightarrow Q = 56$$

$$56 + 19 = 75 \text{ (Not 70)} \Rightarrow R = 75$$

$$75 + 23 = 98$$

So, $Q = 56$ and $R = 75$

Quantity I: 75

Quantity II: 56 and 75

Hence, Quantity I $\geq$ Quantity II

6. Questions

$$2y^2 - 19y - (5P - 2) = 0$$

One root of the above equation is 11.

So,

$$2 * 11^2 - 19 * 11 - (5P - 2) = 0$$

$$242 - 209 - (5P - 2) = 0$$

$$33 = (5P - 2)$$

$$5P = 35$$

$$P = 7$$

Now,

$$2y^2 - 19y - 33 = 0$$

$$2y^2 - 22y + 3y - 33 = 0$$

$$2y(y - 11) + 3(y - 11) = 0$$

$$(y - 11)(2y + 3) = 0$$

$$y = 11 \text{ and } -3/2$$

$$(a + b)x^2 + ax + (b + 19) = 0$$

The coefficient of 'x' and the constant term are equal, and also the coefficient of x^2 in equation I is 'P'.

$$(a + b) = P = 7 \dots\dots\dots (1)$$

$$a = b + 19$$

$$a - b = 19 \dots\dots\dots (2)$$

From (1) and (2):

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$$a = (7 + 19)/2 = 13$$

$$b = (7 - 19)/2 = -6$$

Now,

$$7x^2 + 13x + 13 = 0$$

$$x = [-b \pm \sqrt{(b^2 - 4ac)}]/2a$$

In this formula we taken $a = 7$, $b = 13$, and $c = 13$

$$x = [-13 \pm \sqrt{(13^2 - 4 * 7 * 13)}]/(2 * 7)$$

$$= [-13 \pm \sqrt{(-195)}]/14$$

So, we can find the imaginary root only

(or)

$$D = b^2 - 4ac$$

$$= 13^2 - 4 * 7 * 13$$

$$= 169 - 364$$

$$= -195$$

Since, discriminant of the above equation is negative.

Which means the roots of the equation I will be imaginary.

Answer: D

Hence, relationship between 'x' and 'y' cannot be determined.

7. Questions

$$2y^2 - 19y - (5P - 2) = 0$$

One root of the above equation is 11.

So,

$$2 * 11^2 - 19 * 11 - (5P - 2) = 0$$

$$242 - 209 - (5P - 2) = 0$$

$$33 = (5P - 2)$$

$$5P = 35$$

$$P = 7$$

Now,

$$2y^2 - 19y - 33 = 0$$

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$$2y^2 - 22y + 3y - 33 = 0$$

$$2y(y - 11) + 3(y - 11) = 0$$

$$(y - 11)(2y + 3) = 0$$

$$y = 11 \text{ and } -3/2$$

$$(a + b)x^2 + ax + (b + 19) = 0$$

The coefficient of 'x' and the constant term are equal, and also the coefficient of x^2 in equation I is 'P'.

$$(a + b) = P = 7 \dots\dots\dots (1)$$

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$$= 169 - 364$$

$$= -195$$

Since, discriminant of the above equation is negative.

Which means the roots of the equation I will be imaginary.

Answer: A

$$\text{Equation I: } 7x^2 + 13x + 13 = 0$$

$$\text{A: } 7x^2 + 13x + 13 + 11x - 4 = 0$$

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$$7x^2 + 24x + 9 = 0$$

$$7x^2 + 21x + 3x + 9 = 0$$

$$7x(x + 3) + 3(x + 3) = 0$$

$$(7x + 3)(x + 3) = 0$$

$$x = -3/7 \text{ and } -3$$

This is satisfied.

(or)

$$7x^2 + 24x + 9 = 0$$

$$D = b^2 - 4ac \Rightarrow 24^2 - 4 * 7 * 9 \Rightarrow 576 - 252 = 324 \text{ [Which means roots will be real.]}$$

$$B: 7x^2 + 13x + 13 + x^2 - 11x - 16 = 0$$

$$8x^2 + 2x - 3 = 0$$

$$8x^2 + 6x - 4x - 3 = 0$$

$$2x(4x + 3) - 1(4x + 3) = 0$$

$$(4x + 3)(2x - 1) = 0$$

$$x = -3/4 \text{ and } 1/2$$

This is satisfied.

(or)

$$D = b^2 - 4ac \Rightarrow 2^2 - 4 * 8 * (-3) \Rightarrow 4 + 96 = 100 \text{ [Which means roots will be real.]}$$

$$C: 7x^2 + 13x + 13 + x + 3 = 0$$

$$7x^2 + 14x + 16 = 0$$

$$D = b^2 - 4ac \Rightarrow 14^2 - 4 * 7 * 16 \Rightarrow 196 - 448 = -252 \text{ [Which means roots will be imaginary.]}$$

Hence, only statement A and B are correct.

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8. Questions

$$P + Q + R = 10$$

All P, Q, and R are prime numbers out of which P is the smallest and R is the largest prime number.

Since the sum of three numbers is 10, which means all three numbers must be single-digit numbers. Let's start from the largest single-digit prime number as the value for R.

When R = 7, we cannot find the valid prime numbers as value for P and Q.

When R = 5, sum of P and Q must also be 5. P = 2 and Q = 3

So, $P = 2$, $Q = 3$, and $R = 5$

I: $Px^2 + 13x - 7 = 0$

$$2x^2 + 13x - 7 = 0$$

$$2x^2 + 14x - x - 7 = 0$$

$$2x(x + 7) - 1(x + 7) = 0$$

$$(x + 7)(2x - 1) = 0$$

$$x = -7 \text{ and } 1/2$$

II: $y^2 - Qy - 88 = 0$

$$y^2 - 3y - 88 = 0$$

$$y^2 - 11y + 8y - 88 = 0$$

$$y(y - 11) + 8(y - 11) = 0$$

$$(y - 11)(y + 8) = 0$$

$$y = 11 \text{ and } -8$$

III: $Rz^2 + 24z - R = 0$

$$5z^2 + 24z - 5 = 0$$

$$5z^2 + 25z - z - 5 = 0$$

$$5z(z + 5) - 1(z + 5) = 0$$

$$(z + 5)(5z - 1) = 0$$

$$z = -5 \text{ and } 1/5$$

Answer: D

Largest root of equation I = $1/2$

Largest root of equation II = 11

Largest root of equation III = $1/5$

Required product = $1/2 * 11 * 1/5$

$$= 11/10$$

$$= 1(1/10)$$

9. Questions

$$P + Q + R = 10$$

All P, Q, and R are prime numbers out of which P is the smallest and R is the largest prime number.

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Since the sum of three numbers is 10, which means all three numbers must be single-digit numbers. Let's start from the largest single-digit prime number as the value for R.

When $R = 7$, we cannot find the valid prime numbers as value for P and Q.

When $R = 5$, sum of P and Q must also be 5. $P = 2$ and $Q = 3$

So, $P = 2$, $Q = 3$, and $R = 5$

I: $Px^2 + 13x - 7 = 0$

$$2x^2 + 13x - 7 = 0$$

$$2x^2 + 14x - x - 7 = 0$$

$$2x(x + 7) - 1(x + 7) = 0$$

$$(x + 7)(2x - 1) = 0$$

$$x = -7 \text{ and } 1/2$$

II: $y^2 - Qy - 88 = 0$

$$y^2 - 3y - 88 = 0$$

$$y^2 - 11y + 8y - 88 = 0$$

$$y(y - 11) + 8(y - 11) = 0$$

$$(y - 11)(y + 8) = 0$$

$$y = 11 \text{ and } -8$$

III: $Rz^2 + 24z - R = 0$

$$5z^2 + 24z - 5 = 0$$

$$5z^2 + 25z - z - 5 = 0$$

$$5z(z + 5) - 1(z + 5) = 0$$

$$(z + 5)(5z - 1) = 0$$

$$z = -5 \text{ and } 1/5$$

Answer: B

$$P * Q * R = 2 * 3 * 5 = 30$$

$$4P + 2Q - 1 = 8 + 6 - 1 = 13$$

$$5(Q + R) = 5(3 + 5) = 40$$

Required LCM of 30, 15, and 40 = 1560

10. Questions

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$$P + Q + R = 10$$

All P, Q, and R are prime numbers out of which P is the smallest and R is the largest prime number.

Since the sum of three numbers is 10, which means all three numbers must be single-digit numbers. Let's start from the largest single-digit prime number as the value for R.

When $R = 7$, we cannot find the valid prime numbers as value for P and Q.

When $R = 5$, sum of P and Q must also be 5. $P = 2$ and $Q = 3$

So, $P = 2$, $Q = 3$, and $R = 5$

I: $Px^2 + 13x - 7 = 0$

$$2x^2 + 13x - 7 = 0$$

$$2x^2 + 14x - x - 7 = 0$$

$$2x(x + 7) - 1(x + 7) = 0$$

$$(x + 7)(2x - 1) = 0$$

$$x = -7 \text{ and } 1/2$$

II: $y^2 - Qy - 88 = 0$

$$y^2 - 3y - 88 = 0$$

$$y^2 - 11y + 8y - 88 = 0$$

$$y(y - 11) + 8(y - 11) = 0$$

$$(y - 11)(y + 8) = 0$$

$$y = 11 \text{ and } -8$$

III: $Rz^2 + 24z - R = 0$

$$5z^2 + 24z - 5 = 0$$

$$5z^2 + 25z - z - 5 = 0$$

$$5z(z + 5) - 1(z + 5) = 0$$

$$(z + 5)(5z - 1) = 0$$

$$z = -5 \text{ and } 1/5$$

Answer: E

$$(a - P)(b - Q)(c - R) = 1001$$

$$P = 2, Q = 3, \text{ and } R = 5$$

$$(a - 2)(b - 3)(c - 5) = 1001$$

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1001 can be written as: $7 * 11 * 13$ [We cannot take one factor of 1001 as 1 because it can make value of 'a' as 3.]

After comparing

$$a - 2 = 7 \Rightarrow a = 9$$

$$b - 3 = 11 \Rightarrow b = 14$$

$$c - 5 = 13 \Rightarrow c = 18$$

We can take values in any order to find the sum of values of a, b, and c as it will not change the sum. But we cannot find the exact values of a, b, and c.

$$\text{Required sum} = 9 + 14 + 18$$

$$= 41$$

11. Questions

Answer: E

Let runs scored by Rohit and Rahul in the match is 'x' and '100 - x' respectively.

Let balls faced by Rohit and Rahul in the match 'y' and 'y + 60' respectively.

$$\text{Strike rate of Rohit} = x/y$$

$$\text{Strike rate of Rahul} = (100 - x)/(y + 60)$$

According to the question:

$$x/y = 15/4 * (100 - x)/(y + 60)$$

$$(y + 60)/y = 15/4 * (100 - x)/x$$

From statement I:

$$x = 30 \text{ and } 100 - x = 70$$

$$(y + 60)/y = 15/4 * 70/30$$

After solving: $y = 240/31$ [Not an integer strike rate.]

From statement II:

$$x = 60 \text{ and } 100 - x = 40$$

$$(y + 60)/y = 15/4 * 40/60$$

After solving: $y = 40$ [Integer strike rate.]

From statement III:

$$x = 50 \text{ and } 100 - x = 50$$

$$(y + 60)/y = 15/4 * 50/50$$

After solving: $y = 240/11$ [Not an integer strike rate.]

Hence, only possible number of runs scored by Rohit and Rahul can be 60 and 40 respectively.

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So, only statement II is correct.

12. Questions

Answer: C

Let speed of boat in still water and speed of stream is '3x' km/h and 'x' km/h respectively.

Upstream speed of boat = $3x - x = 2x$ km/h

Downstream speed of boat = $3x + x = 4x$ km/h

According to the question:

$$P/2x = 4$$

$$P = 8x$$

$$Q/4x = 2.5$$

$$Q = 10x$$

$$R/2x + R/4x = 4.5$$

$$(R + 2R)/4x = 4.5$$

$$3R = 18x$$

$$R = 6x$$

Ratio of values of P, Q, and R = P: Q: R = $8x: 10x: 6x = 4: 5: 3$

Which means the correct order of values must be in the ratio of 4: 5: 3.

So, value of P must be 40, Q must be 50, and R must be 30.

Since value of R is correct, so, only values of P and Q must be interchanged.

Hence, correct order is: Q-P-R.

13. Questions

Answer: C

Pages printed by printer A in 1 min = $120/7.5 = 16$ pages

Pages printed by printer B in 1 min = $90/4.5 = 20$ pages

Since, efficiency of printer C is in between the efficiencies of printers A and B. Which means the number of pages printed by printer C in 1 min lies between 16 pages and 20 pages.

Also, efficiency of printer D is less than printer A. Which means number of pages printed by printer D is less than 16 pages.

Possible number of pages printed by printer C in 1 min = 17 or 18 or 19.

Possible number of pages printed by printer D in 1 min = 1 or 2 or 3 or 15.

Minimum number of pages printed by printers C and D together in 1 min = $17 + 1 = 18$ pages

Maximum number of pages printed by printers C and D together in 1 min = $19 + 15 = 34$

So, $34 - 18 = 16$.

14. Questions

Answer: D

Let principal amount is 'P' rupees and rate of interest is R%.

Simple interest at the end of 2 years = $(P * R * 2)/100 = 2PR/100$

Compound interest at the end of 2 years when calculated annually = $P[(1 + R/100)^2 - 1]$

Compound interest at the end of 2 years when rate of interest becomes twice = $P[(1 + 2R/100)^2 - 1]$

According to the question:

$$P[(1 + R/100)^2 - 1] - 2PR/100 = 400$$

$$P[(1 + R^2/100^2 + 2R/100 - 1 - 2R/100)] = 400$$

$$P[R^2/100^2] = 400$$

$$PR^2 = 4 * 10^6 \dots\dots\dots(1)$$

And,

$$P[(1 + 2R/100)^2 - 1] - 2PR/100 = 5600$$

$$P[1 + 4R^2/100^2 + 4R/100 - 1 - 2R/100] = 5600$$

$$P[4R^2/100^2 + 2R/100] = 5600 \dots\dots\dots(2)$$

Equation (1) ÷ Equation (2):

$$PR^2/[P\{4R^2/100^2 + 2R/100\}] = (4 * 10^6)/5600$$

$$R/[\{4R/100^2 + 2/100\}] = (4 * 10^6)/5600$$

$$R/[\{R/2500 + 1/50\}] = (4 * 10^6)/5600$$

$$5600R = 1600R + 80000$$

$$4000R = 80000$$

$$R = 20$$

From equation (1):

$$P * 20^2 = 4 * 10^6$$

$$P = 4000000/400$$

$$P = ₹10000$$

15. Questions

@QuizByMickey

Answer: A

Since only difference between the speeds of car and bike is given. It is not given which between car and bike is faster. So, we cannot individually determine the speeds of both the vehicles but we can calculate the possible speeds of car and bike.

Let the speed of the car and bike be 'x' km/h and 'x - 15' km/h respectively in any order.

Relative speed of car and bike when travelling in opposite direction = $x + (x - 15) = (2x - 15)$ km/h

Distance between them in 1.2 hours = $1.2 * (2x - 15) = 126$

$$2x - 15 = 105$$

$$2x = 120$$

$$x = 60 \text{ km/h}$$

$$x - 15 = 45 \text{ km/h}$$

So, the speed of the car can be 60 km/h or 45 km/h.

the speed of the bike can be 60 km/h or 45 km/h

Statement I: The distance travelled by car in 2 hours can be either 120 km or 90 km but it can never be 100 km.

So, statement I is not correct.

Statement II: Relative speed of car and bike when travelling in the same direction = Difference between their speeds = 15 km/h

Time after which car can catch the bike = $12/15 * 60 = 48$ min

So, statement II is correct.

Statement III: Distance travelled by bike in 1.5 hours can be either 67.5 km or 90 km respectively.

So, statement III is not correct.

Hence, only statement II is correct.

16. Questions

Let the total number of passengers in trains A, B, C, D, and E be 96x, 40x, 56x, 104x, and 64x respectively.

So, the number of female passengers in train B = $40x * (3/5) = 24x$

And the number of female passengers in train C = $56x * (24/35) = 38.4x$

Since, the difference between the number of female passengers in trains B and C = 45

So,

$$38.4x - 24x = 45$$

$$x = 3.125$$

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Train	Total no. of passengers	No. of male passengers	No. of female passengers
A	$96 * 3.125 = 300$	$300 * (7/12) = 175$	$300 - 175 = 125$
B	$40 * 3.125 = 125$	$125 * (2/5) = 50$	$125 - 50 = 75$
C	$56 * 3.125 = 175$	$175 * (11/35) = 55$	$175 - 55 = 120$
D	$104 * 3.125 = 325$	$325 * (9/13) = 225$	$325 - 225 = 100$
E	$64 * 3.125 = 200$	$200 * (3/8) = 75$	$200 - 75 = 125$

Answer: D

Since, the number of male passengers in train C = 55

So, the number of male adults in train C = $55 * (8/11) = 40$

And the number of male children in train C = $55 - 40 = 15$

Since, the number of female passengers in train C = 120

So, the number of female adults in train C = $120 * (13/24) = 65$

And the number of female children in train C = $120 - 65 = 55$

Now, the ratio of the total number of adults to the total number of children in train C = M: N = $(40 + 65): (15 + 55) = 105: 70$

M: N = 3: 2

From option (a):

$$5M = 11N - 5$$

$$5 * 3 = 11 * 2 - 5$$

$$15 \neq 17$$

So, option (a) is not correct.

From option (b):

$$7M - 4N = 13$$

$$7 * 3 - 4 * 2 = 13$$

$$21 - 8 = 13$$

$$13 = 13$$

So, option (b) is correct.

From (c):

$$M^3 - N^3 = 21$$

$$3^3 - 2^3 = 21$$

$$27 - 8 = 21$$

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19 ≠ 21

So, option (c) is not correct.

Hence, both options (a) and (c) are not correct.

17. Questions

Let the total number of passengers in trains A, B, C, D, and E be $96x$, $40x$, $56x$, $104x$, and $64x$ respectively.

So, the number of female passengers in train B = $40x * (3/5) = 24x$

And the number of female passengers in train C = $56x * (24/35) = 38.4x$

Since, the difference between the number of female passengers in trains B and C = 45

So,

$$38.4x - 24x = 45$$

$$x = 3.125$$

Train	Total no. of passengers	No. of male passengers	No. of female passengers
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C	$56 * 3.125 = 175$	$175 * (11/35) = 55$	$175 - 55 = 120$
D	$104 * 3.125 = 325$	$325 * (9/13) = 225$	$325 - 225 = 100$
E	$64 * 3.125 = 200$	$200 * (3/8) = 75$	$200 - 75 = 125$

Answer: B

Since, the number of male passengers in train E = 75

So, the number of male adults in train E = $75 * (11/15) = 55$

And the number of male children in train E = $75 - 55 = 20$

Since, the number of female passengers in train E = 125

So, the number of female adults in train E = $125 * (18/25) = 90$

And the number of female children in train E = $125 - 90 = 35$ @QuizByMickey

Since, the total amount received by selling tickets for adults in train E = $(55 + 90) * 80 = ₹11600$

And the total amount received by selling tickets for children in train E = $(20 + 35) * 40 = ₹2200$

So, the required difference = $[(11600 - 2200)/2200] * 100 = 427\%$ (approx.)

18. Questions

Let the total number of passengers in trains A, B, C, D, and E be $96x$, $40x$, $56x$, $104x$, and $64x$ respectively.

So, the number of female passengers in train B = $40x * (3/5) = 24x$

And the number of female passengers in train C = $56x * (24/35) = 38.4x$

Since, the difference between the number of female passengers in trains B and C = 45

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Train	Total no. of passengers	No. of male passengers	No. of female passengers
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E	$64 * 3.125 = 200$	$200 * (3/8) = 75$	$200 - 75 = 125$

Answer: C

Since, the total number of male passengers in all five trains together = $175 + 50 + 55 + 225 + 75 = 580$

So, the degree distribution of the number of male passengers in train B:

$$P = 360^\circ * (50/580)$$

$$P = (900/29)^\circ$$

Since the total number of female passengers in all five trains together = $125 + 75 + 120 + 100 + 125 = 545$

So, the degree distribution of the number of female passengers in train D:

$$Q = 360^\circ * (100/545)$$

$$Q = (7200/109)^\circ$$

$$\text{Required ratio} = P: Q = (900/29): (7200/109) = (1/29): (8/109) = 109: 232$$

19. Questions

Let the total number of passengers in trains A, B, C, D, and E be $96x$, $40x$, $56x$, $104x$, and $64x$ respectively.

So, the number of female passengers in train B = $40x * (3/5) = 24x$

And the number of female passengers in train C = $56x * (24/35) = 38.4x$

Since, the difference between the number of female passengers in trains B and C = 45

So,

$$38.4x - 24x = 45$$

$$x = 3.125$$

Train	Total no. of passengers	No. of male passengers	No. of female passengers
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E	$64 * 3.125 = 200$	$200 * (3/8) = 75$	$200 - 75 = 125$

Answer: B

Since, the total number of passengers in train A = 300

So, the total number of children in train A = $41\frac{2}{3}\%$ of 300 = 125

The number of male children in train A = $125 * (12/25) = 60$

And the number of female children in train A = $125 - 60 = 65$

Since, the number of male passengers in train A = 175

So, the number of male adults in train A = $175 - 60 = 115$

Since, the number of female passengers in train A = 125

So, the number of female adults in train A = $125 - 65 = 60$

Required ratio = 115: 60 = 23: 12

20. Questions

Let the total number of passengers in trains A, B, C, D, and E be 96x, 40x, 56x, 104x, and 64x respectively.

So, the number of female passengers in train B = $40x * (3/5) = 24x$

And the number of female passengers in train C = $56x * (24/35) = 38.4x$

Since, the difference between the number of female passengers in trains B and C = 45

So,

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$$x = 3.125$$

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E	$64 * 3.125 = 200$	$200 * (3/8) = 75$	$200 - 75 = 125$

Answer: E

1st term of the series = 50

2nd term of the series = 55

3rd term of the series = 75

4th term of the series = 55 + 100 = 155

5th term of the series = 325 + 150 = 475

Logic in the series:

$$50 + 5 * 4^0 = 55$$

$$55 + 5 * 4^1 = 75$$

$$75 + 5 * 4^2 = 155$$

$$155 + 5 * 4^3 = 475$$

$$\text{So, the 6th term of the series} = 475 + 5 * 4^4 = 1755$$

(or)

50	55	75	155	475	1755
	+5	+20	+80	+320	+1280
		*4	*4	*4	*4

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